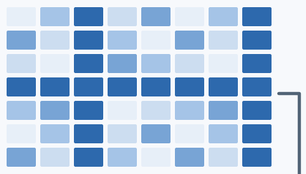
FastVCL Sparse-to-tube video copy localization

PROPOSED METHOD · SCHEMATIC

1 Fixed features

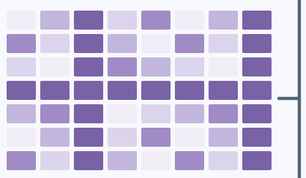
Query A / reference B

Query A



F_A: M × 256

Reference B

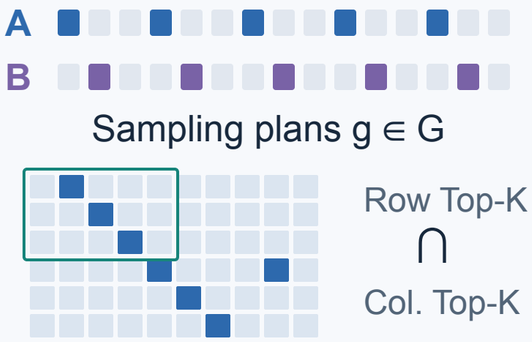


F_B: N × 256

L2 normalization
+ validity masks

2 Sparse matching

Sample → compare → retain

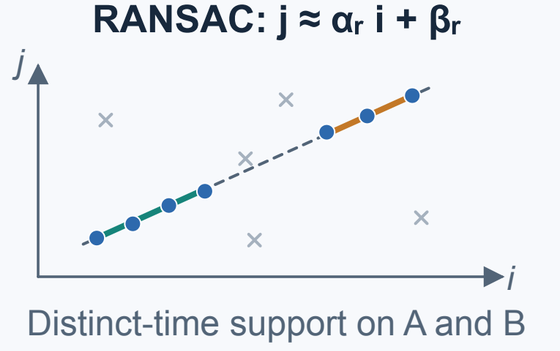


Deduplicate coordinates; cap

Anchor set {(i, j, S_{ij})}

3 Temporal proposals

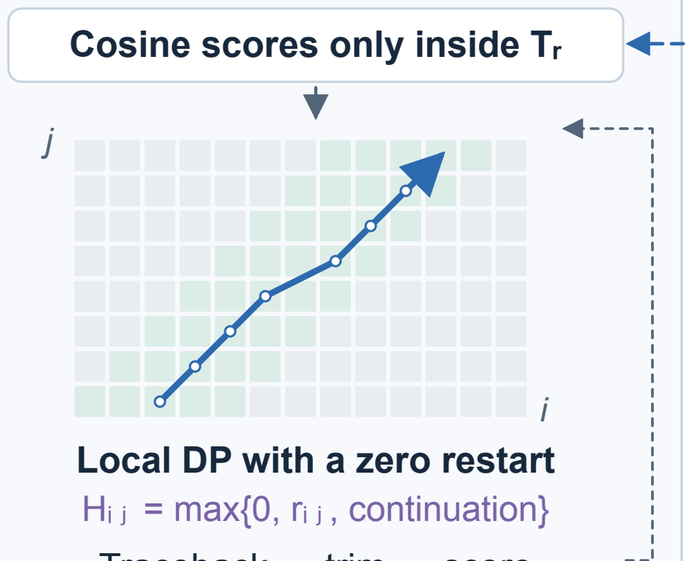
Fit → validate → split



Finite T_r: $\ell_r \le i < u_r$

4 Local refinement

Align → recover → candidates



Candidate intervals + matched indices

ADDITIONAL FRAME INPUT

PROPOSED VERIFICATION EXTENSION

A/B frame context



Index → frame mapping
Original video frames

Select evidence

Matched frame pairs / snippets

Budgeted verification calls

Selection policy: to be defined

5 seltive model

Visual copy verification

COPY / NOT_COPY / UNCERTAIN

Model backend: to be specified

Final copy intervals

Fuse verification evidence

Uncertain / failed: retain core result

Y = {(a_s, b_s, a_e, b_e, q)}

Fusion policy: to be defined

Unselected candidates retain the core result

Full-resolution normalized descriptors + validity masks

● anchor — track ▭ tube → local path

Core + proposed selective verification · feature-index coordinates